

Администрирование сетевых подсистем

Лабораторная работа №9

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Цели и задачи работы

Приобретение практических навыков по установке и конфигурированию POP3/IMAP-сервера с использованием Dovecot и Postfix.

Выполнение лабораторной работы

Установка и настройка Dovecot

```
root@server:~# sudo -i
dovecot.conf [----] 21 L: [ 1*24 25/103] *(1177/4317b) 0010 0x00A [*)(X
## Dovecot configuration file

# If you're in a hurry, see http://wiki2.dovecot.org/QuickConfiguration

# "doveconf -n" command gives a clean output of the changed settings. Use it
# instead of copy&pasting files when posting to the Dovecot mailing list.

# "#" character and everything after it is treated as comments. Extra spaces
# and tabs are ignored. If you want to use either of these explicitly, put the
# value inside quotes, eg.: key = "# char and trailing whitespace "

# Most (but not all) settings can be overridden by different protocols and/or
# source/destination IPs by placing the settings inside sections, for example:
# protocol imap { }, local 127.0.0.1 { }, remote 10.0.0.0/8 { }

# Default values are shown for each setting, it's not required to uncomment
# those. These are exceptions to this though: No sections (e.g. namespace {})
# or plugin settings are added by default, they're listed only as examples.
# Paths are also just examples with the real defaults listed based on configure
# options. The paths listed here are for configure --prefix=/usr
# --sysconfdir=/etc --localstatedir=/var

# Protocols we want to be serving.

protocols = imap pop3

# A comma separated list of IPs or hosts where to listen in for connections.
# "" listens in all IPv4 interfaces, "::" listens in all IPv6 interfaces.
# If you want to specify non-default ports or anything more complex,
# edit conf.d/master.conf.
#listen = *, ::

# Base directory where to store runtime data.
#base_dir = /var/run/dovecot/

# Name of this instance. In multi-instance setup doveadm and other commands
```

Конфигурация основных протоколов

```
root@server:~# sudo -i
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# --sysconfdir=/etc --localstatedir=/var

# Protocols we want to be serving.

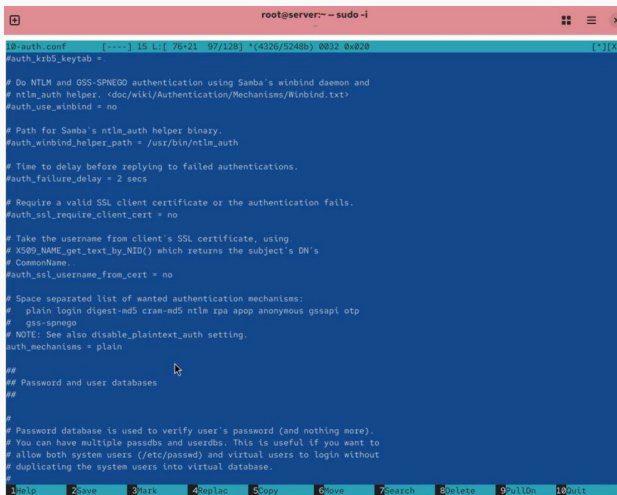
protocols = imap pop3

# A comma separated list of IPs or hosts where to listen in for connections.
# "*" listens in all IPv4 interfaces, "::" listens in all IPv6 interfaces.
# If you want to specify non-default ports or anything more complex,
# edit conf.d/master.conf.
#listen = *, ::

# Base directory where to store runtime data.
#base_dir = /var/run/dovecot/

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```

Настройка аутентификации



```
root@server:~ - sudo -i
10-auth.conf [----] 15 L: [ 76*21 97/128 ] *(4326/5248b) 0032 0x020 (*)[X]
#auth_krb5_keytab =

# Do NTLM and GSS-SPNEGO authentication using Samba's winbind daemon and
# ntlm_auth helper. <doc/wiki/Authentication/Mechanisms/Winbind.txt>
#auth_use_winbind = no

# Path for Samba's ntlm_auth helper binary.
#auth_winbind_helper_path = /usr/bin/ntlm_auth

# Time to delay before replying to failed authentications.
#auth_failure_delay = 2 secs

# Require a valid SSL client certificate or the authentication fails.
#auth_ssl_require_client_cert = no

# Take the username from client's SSL certificate, using
# X509_NAME_get_text_by_NID() which returns the subject's DN's
# CommonName.
#auth_ssl_username_from_cert = no

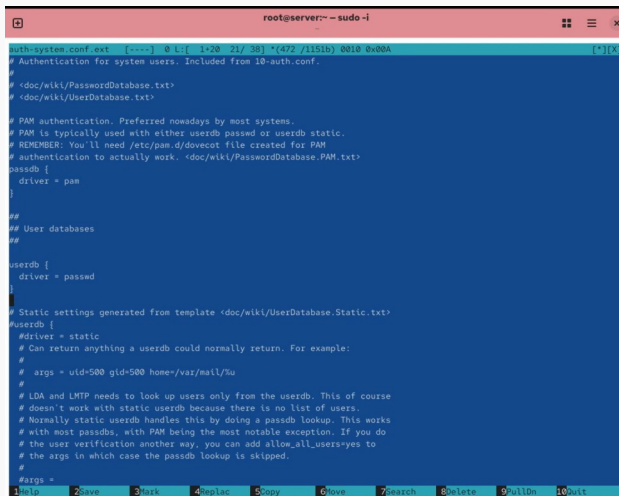
# Space separated list of wanted authentication mechanisms:
# plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp
# gss-spnego
# NOTE: See also disable_plaintext_auth setting.
auth_mechanisms = plain

##
## Password and user databases
##

# Password database is used to verify user's password (and nothing more).
# You can have multiple passwd and userdb. This is useful if you want to
# allow both system users (/etc/passwd) and virtual users to login without
# duplicating the system users into virtual database.
#
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullOn 10Quit
```

Рис. 3: Настройка метода аутентификации

Настройка источников пользователей



```
auth-system.conf.ext [----] 0 L:[ 1+20 21/ 38] *(472 /1151b) 0010 0x00A [*][X]
# Authentication for system users. Included from 10-auth.conf.
#
# <doc/wiki/PasswordDatabase.txt>
# <doc/wiki/UserDatabase.txt>
#
# PAM authentication. Preferred nowadays by most systems.
# PAM is typically used with either userdb passwd or userdb static.
# REMEMBER: You'll need /etc/pam.d/dovecot file created for PAM
# authentication to actually work. <doc/wiki/PasswordDatabase.PAM.txt>
passdb {
    driver = pam
}

##
## User databases
##
userdb {
    driver = passwd
}

# Static settings generated from template <doc/wiki/UserDatabase.Static.txt>
#userdb {
#    #driver = static
#    # Can return anything a userdb could normally return. For example:
#    #
#    # args = uid=500 gid=500 home=/var/mail/%u
#    #
#    # LDA and LMTP needs to look up users only from the userdb. This of course
#    # doesn't work with static userdb because there is no list of users.
#    # Normally static userdb handles this by doing a passdb lookup. This works
#    # with most passdbs, with PAM being the most notable exception. If you do
#    # the user verification another way, you can add allow_all_users=yes to
#    # the args in which case the passdb lookup is skipped.
#    #
#    #args =
#}

1 Help 2 Save 3 Mark 4 Replac 5 Copy 6 Move 7 Search 8 Delete 9 PullDn 10 Quit
```

Рис. 4: Файл auth-system.conf.ext

Указание расположения почтовых ящиков

```
root@server:~# sudo -i
10-mail.conf [----] 0 L:[ 1+23 24/419] *(852 /17793b) 0010 0x00A [*][X]
##
## Mailbox locations and namespaces
##
# Location for users' mailboxes. The default is empty, which means that Dovecot
# tries to find the mailboxes automatically. This won't work if the user
# doesn't yet have any mail, so you should explicitly tell Dovecot the full
# location.
#
# If you're using mbox, giving a path to the INBOX file (eg. /var/mail/%u)
# isn't enough. You'll also need to tell Dovecot where the other mailboxes are
# kept. This is called the "root mail directory", and it must be the first
# path given in the mail_location setting.
#
# There are a few special variables you can use, eg.:
#
# %u - username
# %n - user part in user@domain, same as %u if there's no domain
# %d - domain part in user@domain, empty if there's no domain
# %h - home directory
#
# See doc/wiki/Variables.txt for full list. Some examples:
#
mail_location = maildir:~/Maildir
#
# mail_location = mbox:~/mail:INBOX=/var/mail/%u
# mail_location = mbox:/var/mail/%d/%ln%n:INDEX=/var/indexes/%d/%ln%n
#
# <doc/wiki/MailLocation.txt>
#mail_location =
#
# If you need to set multiple mailbox locations or want to change default
# namespace settings, you can do it by defining namespace sections.
#
1|help 2|Save 3|Mark 4|Replac 5|Copy 6|Move 7|Search 8|Delete 9|PullDn 10|Quit
```

Настройка Postfix и межсетевого экрана

```
[root@server.manovichkov.net ~]# firewall-cmd --add-service=pop3 --permanent
success
[root@server.manovichkov.net ~]# firewall-cmd --add-service=pop3s --permanent
success
[root@server.manovichkov.net ~]# firewall-cmd --add-service=imap --permanent
success
[root@server.manovichkov.net ~]# firewall-cmd --add-service=imaps --permanent
success
[root@server.manovichkov.net ~]# firewall-cmd --reload
success
[root@server.manovichkov.net ~]# firewall-cmd --list-services
cockpit dhcpv6-client dns http https imap imaps pop3 pop3s smtp ssh
[root@server.manovichkov.net ~]# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
[root@server.manovichkov.net ~]# systemctl restart postfix
[root@server.manovichkov.net ~]# systemctl enable dovecot
Created symlink '/etc/systemd/system/multi-user.target.wants/dovecot.service' → '/usr/lib/systemd/system/dovecot.service'.
[root@server.manovichkov.net ~]# systemctl start dovecot
[root@server.manovichkov.net ~]# █
```

Рис. 6: Настройка межсетевого экрана и запуск служб

```
[manovichkov@server.manovichkov.net ~]$ sudo tail -f /var/log/maillog
[sudo] password for manovichkov:
Oct 13 06:37:41 server postfix/postfix-script[1485]: starting the Postfix mail system
Oct 13 06:37:41 server postfix/master[1501]: daemon started -- version 3.8.5, configuration /etc/postfix
Oct 13 06:37:41 server postfix/qmgr[1505]: 19AF1609E75E: from=<>, size=2618, nrcpt=1 (queue active)
Oct 13 06:37:44 server postfix/smtp[1511]: connect to client.manovichkov.net[192.168.1.30]:25: No route to host
Oct 13 06:37:44 server postfix/smtp[1511]: 19AF1609E75E: to=<manovichkov@client.manovichkov.net>, relay=none, delay=418987
, delays=418983/0.03/3.1/0, dsn=4.4.1, status=deferred (connect to client.manovichkov.net[192.168.1.30]:25: No route to ho
st)
Oct 13 06:48:45 server postfix/postfix-script[14670]: stopping the Postfix mail system
Oct 13 06:48:45 server postfix/master[1501]: terminating on signal 15
Oct 13 06:48:46 server postfix/postfix-script[14748]: starting the Postfix mail system
Oct 13 06:48:46 server postfix/master[14750]: daemon started -- version 3.8.5, configuration /etc/postfix
Oct 13 06:49:04 server dovecot[14952]: master: Dovecot v2.3.21 (47349e2482) starting up for imap, pop3
```

Рис. 7: Мониторинг maillog

Настройка почтового клиента Evolution

Cancel

Back

Identity

Next

Welcome

Restore from Backup

Identity

Receiving Email

Sending Email

Account Summary

Done

Please enter your name and email address below. The "optional" fields below do not need to be filled in, unless you wish to include this information in email you send.

Required Information

Full Name:

Email Address:

Optional Information

Reply-To:

Organization:

Signature: None ▾ Add New Signature...

Aliases:

Add

Edit

Remove

☒ Look up mail server details based on the entered e-mail address

Настройка получения почты (IMAP)

Cancel

Back

Receiving Email

Next

Welcome

Restore from Backup

Identity

Receiving Email

Receiving Options

Sending Email

Account Summary

Done

Server Type: IMAP

Description: For reading and storing mail on IMAP servers.

Configuration

Server: mail.manovichkov.net

Port: 993

Username: manovichkov

Security

Encryption method: TLS on a dedicated port

Authentication

Check for Supported Types

Password

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Настройка отправки почты (SMTP)

CancelBack

Sending Email

NextFinish

Welcome
Restore from Backup
Identity
Receiving Email
Receiving Options
Sending Email
Account Summary
Done

Server Type: SMTP

Description: For delivering mail by connecting to a remote mailhub using SMTP.

Configuration

Server: mail.manovichkov.netPort: 25

☐ Server requires authentication

Security

Encryption method: No encryption

Authentication

Type: Check for Supported TypesPLAIN

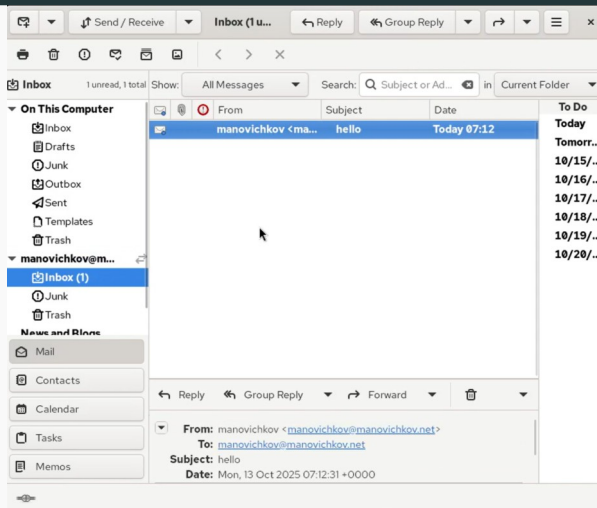
Username: manovichkov

Send Options

☐ Re-encode message before send

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Отправка и получение писем



```
Oct 13 07:12:06 server dovecot[1655]: imap-login: Login: user=<manovichkov>, method=PLAIN, rip=192.168.1.30, lip=192.168.1.1, mpid=10217, TLS, session=<qSMk/ARBTtPAqAEe>
Oct 13 07:12:31 server postfix/smtpd[10268]: connect from client.manovichkov.net[192.168.1.30]
Oct 13 07:12:31 server postfix/smtpd[10268]: 5FFD3609E75D: client=client.manovichkov.net[192.168.1.30]
Oct 13 07:12:31 server postfix/cleanup[10274]: 5FFD3609E75D: message-id=<7940bdcc890df5196afb8ec676ff3fb64d28cba4.came
l@manovichkov.net>
Oct 13 07:12:31 server postfix/smtpd[10268]: disconnect from client.manovichkov.net[192.168.1.30] ehlo=1 mail=1 rcpt=1
data=1 quit=1 commands=5
Oct 13 07:12:31 server postfix/qmgr[1497]: 5FFD3609E75D: from=<manovichkov@manovichkov.net>, size=570, nrcpt=1 (queue
active)
Oct 13 07:12:31 server postfix/local[10275]: 5FFD3609E75D: to=<manovichkov@manovichkov.net>, relay=local, delay=0.01,
delays=0.01/0.01/0/0, dsn=2.0.0, status=sent (delivered to maildir)
Oct 13 07:12:31 server postfix/qmgr[1497]: 5FFD3609E75D: removed
```

Рис. 12: Записи в maillog

Проверка через Telnet

```
root@server:~ - sudo -i
manovichkov@server:~ - sudo tail -f /var/log/maillog
root@server:~ - sudo -i

Return-Path: <manovichkov@manovichkov.net>
X-Original-To: manovichkov@manovichkov.net
Delivered-To: manovichkov@manovichkov.net
Received: from client.manovichkov.net (client.manovichkov.net [192.168.1.30])
    by server.manovichkov.net (Postfix) with ESMTP id 5FFD3609E75D
    for <manovichkov@manovichkov.net>; Mon, 13 Oct 2025 07:12:31 +0000 (UTC)
Message-ID: <7940bdcc890df5196afb8ec676ff3fb64d28cba4.camel@manovichkov.net>
Subject: hello
From: manovichkov <manovichkov@manovichkov.net>
To: manovichkov@manovichkov.net
Date: Mon, 13 Oct 2025 07:12:31 +0000
Content-Type: text/plain
Content-Transfer-Encoding: 7bit
User-Agent: Evolution 3.52.4 (3.52.4-2.el10_0)
MIME-Version: 1.0

hello
.
dele 2
+OK Marked to be deleted.
quit
+OK Logging out, messages deleted.
Connection closed by foreign host.
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]#
```

```
[root@server.manovichkov.net ~]# cd /vagrant/provision/server/
[root@server.manovichkov.net server]# mkdir -p /vagrant/provision/server/mail/etc/dovecot/conf.d
[root@server.manovichkov.net server]# cp -R /etc/dovecot/dovecot.conf /vagrant/provision/server/mail/etc/dovecot/
[root@server.manovichkov.net server]# cp -R /etc/dovecot/conf.d/10-auth.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/
[root@server.manovichkov.net server]# cp -R /etc/dovecot/conf.d/auth-system.conf.ext /vagrant/provision/server/mail/etc/dovecot/conf.d/
[root@server.manovichkov.net server]# cp -R /etc/dovecot/conf.d/10-mail.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/
```

Рис. 14: Копирование конфигурационных файлов Dovecot

Выводы по проделанной работе

В ходе лабораторной работы был установлен и настроен почтовый сервер Dovecot, обеспечивающий работу по протоколам IMAP и POP3 в связке с Postfix.

Реализована аутентификация через системные учётные записи, проверена отправка и получение писем через почтовый клиент и Telnet.

Настроены firewall и SELinux, а конфигурационные файлы сохранены для автоматического развертывания.